SCHOOL OF ENGINERING AND TECHNOLOGY

ASSIGNMENT COVER SHEET

COURSE: NET1014 – Networking Principles

LEVEL: BCNS, BIT, BCS, BSE, BDS - Year 1

ACADEMIC SESSION: February 2025 Semester

DEADLINE: 28th Mar 2025 – 11:59PM

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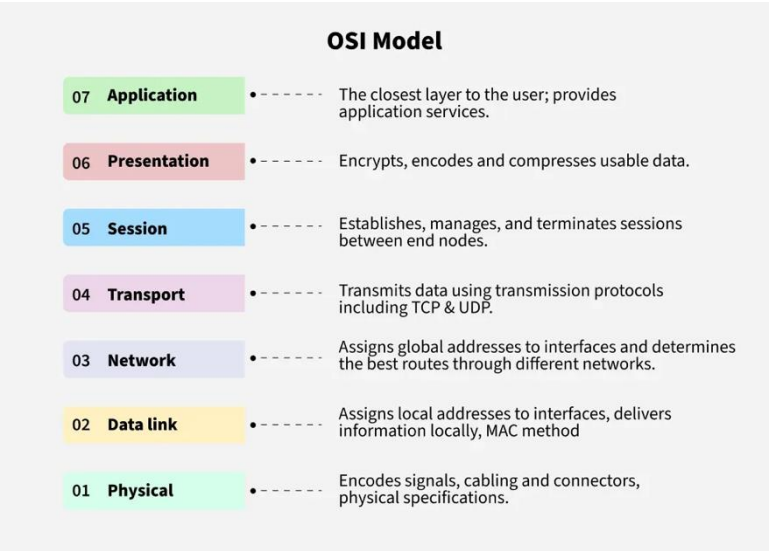
I. Comprehensive Analysis of OSI Protocol Stack

1. Overview of the OSI Model

The OSI (Open Systems Interconnection) model is a conceptual framework used to understand and implement standard protocols in network communications. It divides the network communication process into seven layers, each with specific functions and tasks. The layers are designed to work together to ensure seamless data transmission across networks.

2. The 7 Layers of the OSI Model



Layer 1: Physical Layer

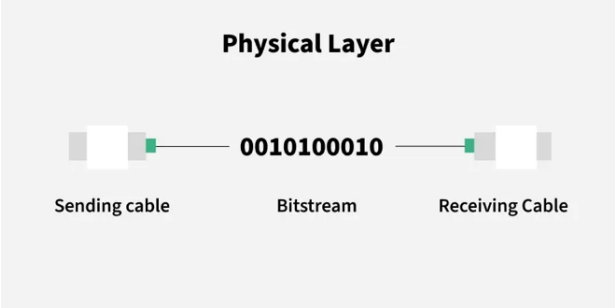
The Physical Layer handles the **transmission of raw data (0s and 1s)** over a physical medium, defining **hardware characteristics** like cables, signals, and connectors to ensure reliable communication between network devices.

Key Responsibilities:

- **Bit Transmission** – Converts data packets into **electrical signals, light pulses, or radio waves** for transmission.
- **Signal Encoding** – Defines how data is modulated (e.g., digital-to-analog conversion).
- **Media Control** – Regulates the use of **wired (Ethernet, fiber optics) and wireless (Wi-Fi, Bluetooth) transmission methods**.
- **Transmission Modes** – Supports **simplex, half-duplex, and full-duplex communication**.

- **Synchronization** – Ensures the sender and receiver are synchronized to correctly interpret bits.
- **Topology Implementation** – Defines **network layout structures** (star, bus, ring, mesh, etc.).
- **Error Detection** – Identifies transmission errors caused by noise or interference but does not correct them.

Protocols/Examples: Ethernet (IEEE 802.3), USB, Bluetooth, DSL.



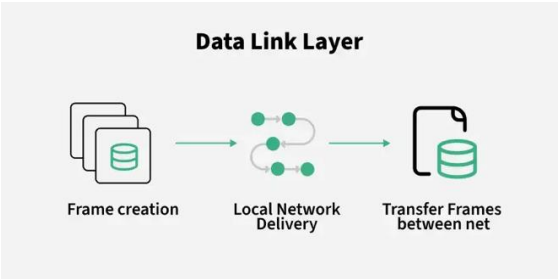
Layer 2: Data Link Layer

The **Data Link Layer** ensures **error-free transmission of data frames between directly connected nodes**. It **controls access to the network medium**, organizes data into **frames**, and assigns **MAC (Media Access Control) addresses** to devices.

Key Responsibilities:

- **Framing** – Divides data into structured **frames** for transmission.
- **MAC Addressing** – Assigns **unique identifiers** to network devices.
- **Error Detection & Correction** – Uses **Cyclic Redundancy Check (CRC)** to detect corrupted frames.
- **Flow Control** – Manages data flow to prevent congestion.
- **Media Access Control (MAC)** – Controls how devices share the network medium.

Protocols/Examples: Ethernet (IEEE 802.3), PPP, Wi-Fi (IEEE 802.11), VLANs.



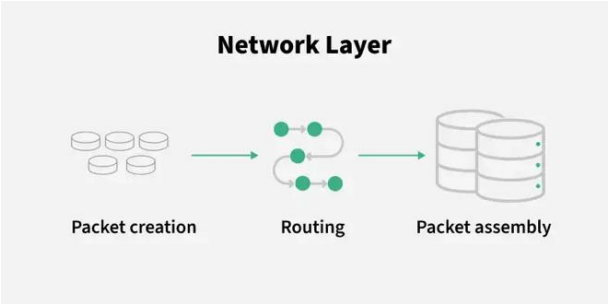
Layer 3: Network Layer

The **Network Layer** is responsible for **logical addressing and routing**. It assigns **IP addresses** to devices and determines the best **path for packet forwarding** across multiple networks.

Key Responsibilities:

- **Logical Addressing** – Uses **IPv4 or IPv6** to identify devices.
- **Routing** – Determines the most efficient path for data transmission.
- **Packet Fragmentation** – Breaks large packets into smaller segments.
- **Network Congestion Control** – Prevents overload on network routers.

Protocols/Examples: IP (IPv4, IPv6), ICMP, ARP, OSPF.



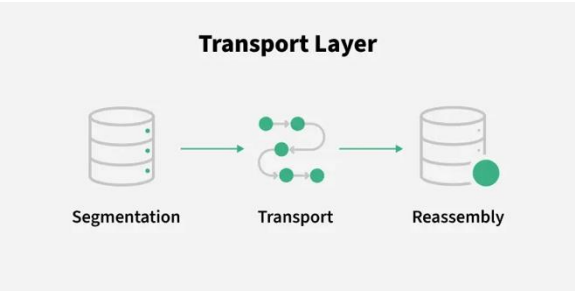
Layer 4: Transport Layer

The **Transport Layer** ensures **end-to-end communication between devices**. It manages **data segmentation, error correction, and flow control**.

Key Responsibilities:

- **Reliable vs. Unreliable Transport** – Supports **connection-oriented (TCP) and connectionless (UDP) communication**.
- **Error Recovery** – Detects **lost or corrupted packets** and requests retransmission.
- **Flow Control** – Adjusts data transmission speed based on network conditions.

Protocols/Examples: TCP (reliable), UDP (unreliable).



Layer 5: Session Layer

The **Session Layer** establishes and manages **communication sessions** between devices.

Key Responsibilities:

- **Session Authentication** – Ensures secure communication.
- **Synchronization** – Maintains session checkpoints to recover lost data.

Protocols/Examples: NetBIOS, RPC, SIP, PPTP.

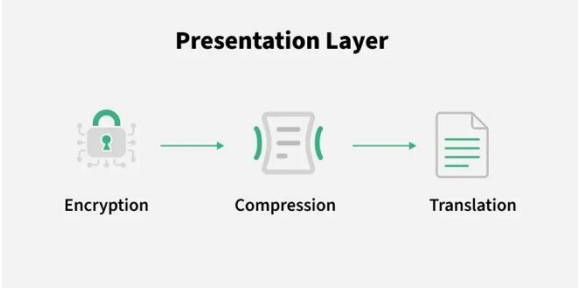
Layer 6: Presentation Layer

The **Presentation Layer** translates, compresses, and encrypts data.

Key Responsibilities:

- **Data Compression** – Reduces file size for faster transmission.
- **Encryption & Decryption** – Secures data in transit.

Protocols/Examples: SSL/TLS, JPEG, MPEG, ASCII.



Layer 7: Application Layer

Provides network services directly to end-user applications for communication and data exchange.

Key Responsibilities:

- **Network Service Access** – Provides interfaces for applications to communicate over a network.
- **Data Exchange & Processing** – Handles data formatting, encoding, and presentation for applications.
- **User Authentication & Authorization** – Ensures secure access to network resources.
- **Protocol Handling** – Manages communication protocols like **HTTP, FTP, DNS, and SMTP**.
- **Error Handling & Logging** – Detects application-level errors and maintains logs for troubleshooting.

Protocols/Examples: HTTP, FTP, SMTP, DNS.

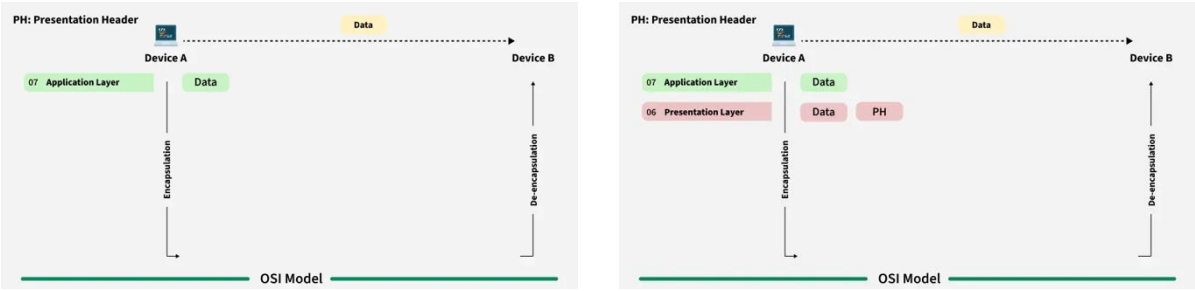
3. Visualizations

- **Inter-layer Relationships:** A diagram showing how data flows from the application layer (Layer 7) down to the physical layer (Layer 1) and vice versa. Each layer adds its own header/trailer to the data packet.
- **Advantages of OSI Model:**
  - Standardization: Provides a universal standard for network communication.
  - Modularity: Each layer operates independently, making troubleshooting easier.
  - Interoperability: Allows different systems to communicate seamlessly.
  - Security and flexibility: Ensure secure, structured data transfer between layers.
- **Challenges of OSI Model:**
  - Complexity: The model can be overly complex for simple networks.
  - Implementation: Not all protocols fit neatly into the OSI layers.
  - Performance overhead: The layered structure can introduce inefficiencies.
  - Not fully adopted: Many modern networks primarily use the TCP/IP model rather than full OSI model.

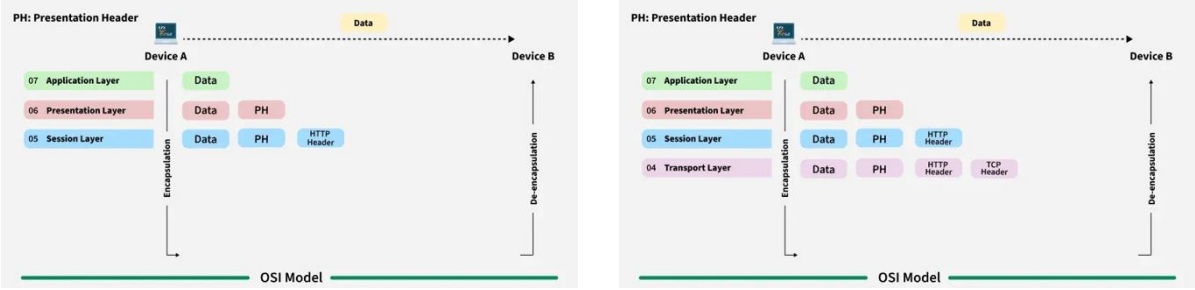
4. Real-World Examples: Accessing a Website Using the OSI Model

When a user on Device A (a laptop) access a website, data flows through the OSI layers, is transmitted over the network and reaches Device B (web server).

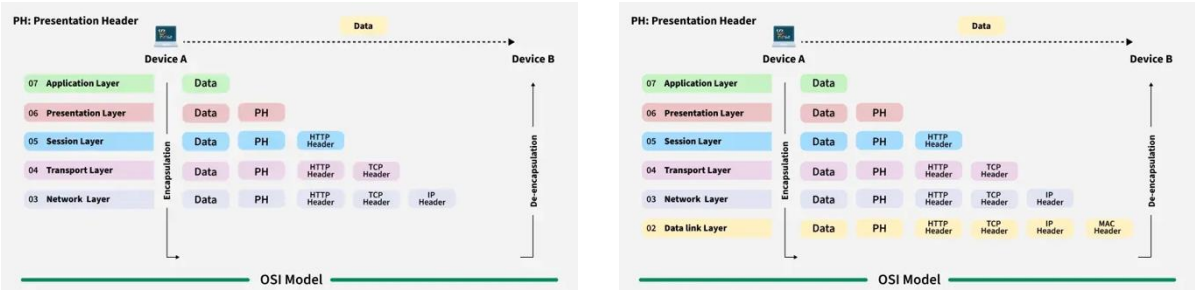
- **Application Layer:** HTTP enables web browsing. The browser sends an HTTP GET request.
- **Presentation Layer:** SSL/TLS encrypts web traffic for secure communication.
- **Session Layer:** SIP manages sessions (e.g., VoIP calls, though HTTP uses TCP sessions).
- **Transport Layer:** TCP ensures reliable file transfers by segmenting the data.
- **Network Layer:** Routers use IP address to route packets toward website.
- **Data Link Layer:** Switch forward data based on MAC address within the local network.
- **Physical Layer:** Ethernet cables (or Wi-Fi signals) physically transmit data as electrical signals or radio waves.



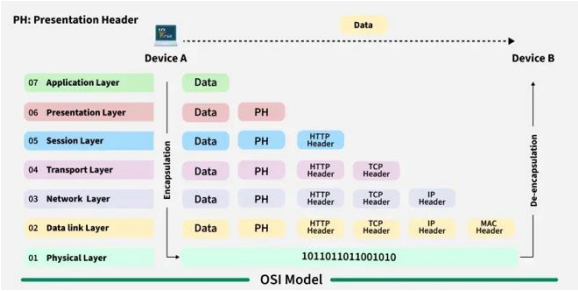
Picture 1 and 2

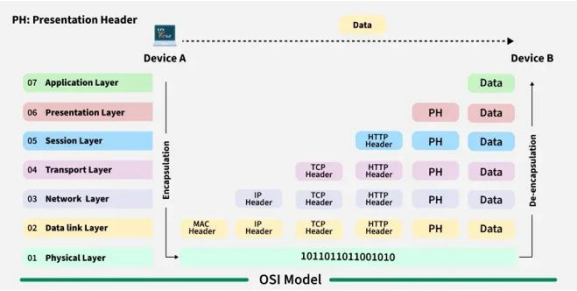


Picture 3 and 4



Picture 5 and 6





Picture 7 and 8

II. Design and Implementation of a Network

Part A: Design

1. List of Required Equipment

- **1 Router**
  - Acts as the main router for the entire network. It is used to connect multiple devices to the internet, including devices from different subnets, for example HUMAC.

- **5 Switches**
  - Each switch contains 24 Ethernet ports, allowing all wired devices in a VLAN to be connected, thus no additional switch in each VLAN is required. It mainly handles data transfer from the connected devices.

- **3 Access Points**
  - They enable wireless connections for devices, especially for devices in Advanced Wireless and IoT Lab. This allows wireless devices to access the internet without the use of cables.

- **Cables**
  - Copper Straight-Through Cables (471 meters): Connects switches to end devices and router, allowing network traffic transmission.
  - Copper Cross-Over Cables (450 meters): It allows connection and communication between two same types of devices, in our case, a switch with another switch.

*End devices such as laptops, PCs, printers, and others are not listed as they are not part of the equipment needed for network functionality.

2. Estimated Cost of Equipment

Core Equipment

Component	Model/ Specification	Quantity	Unit Price (RM)	Total Cost (RM)
Router	Cisco 1941	1	**1200.00**	**1200.00**
Switch	Cisco 2960-24TT	5	**550.00**	**2750.00**
Access Point	Cisco Wireless Access Point	3	**900.00**	**2700.00**
			Total:	**6650.00**

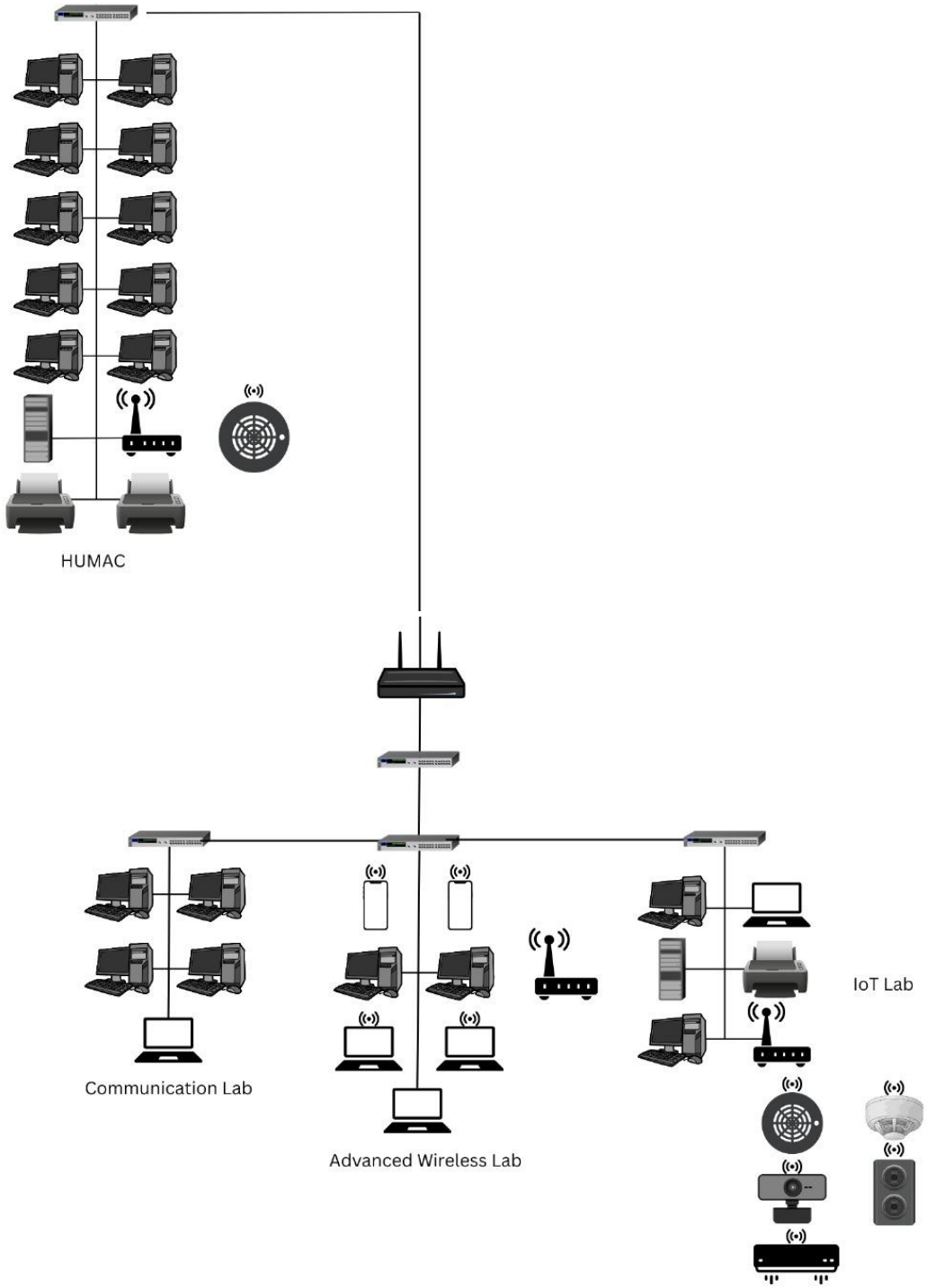
Connectivity Equipment

Cable Type	Lab / Connection	Length (meter)	Cost per meter (RM)	Total Cost (RM)
Copper Straight-Through	HUMAC Lab	120 (10m x 12)	**2.30**	**276.00**
	Communications Lab	50 (10m x 5)		**115.00**
	Advanced Wireless Lab	40 (10m x 4)		**92.00**
	IoT Lab	60 (10m x 6)		**138.00**
	Switch 2 to Router 1	200		**460.00**
	Switch 1 to Router 1	1		**2.30**
Copper Cross-Over	Switch 4 to Switch 1	150	**2.60**	**390.00**
	Switch 3 to Switch 1	150		**390.00**
	Switch 5 to Switch 1	150		**390.00**
		*m = meter	**Total:**	**2253.30**

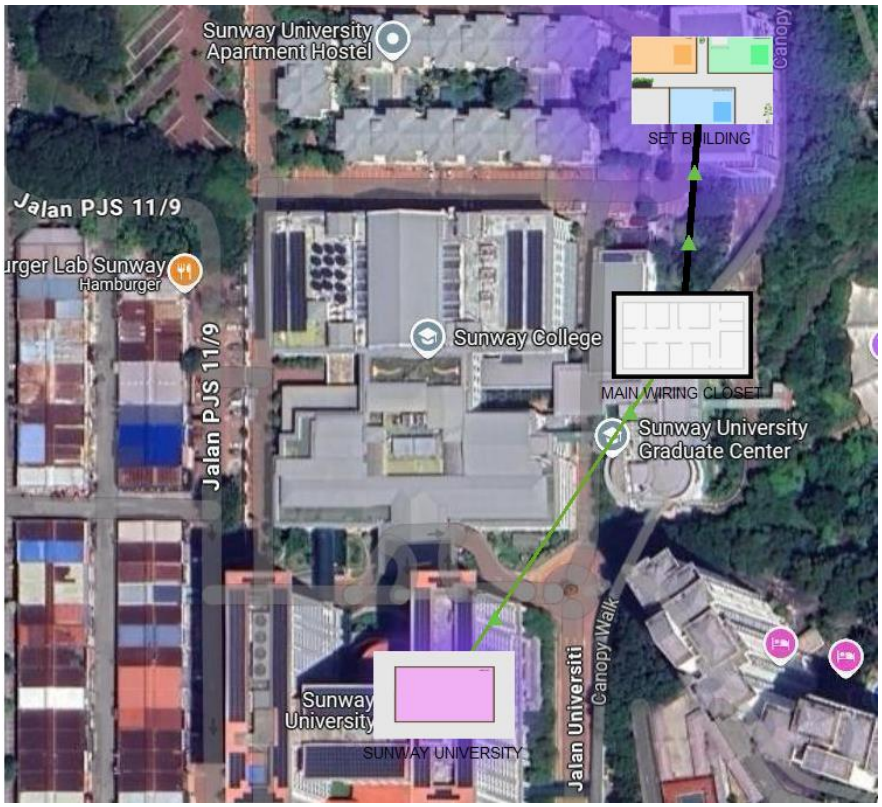
Total Estimated Cost of Equipment

Category	Total Cost (RM)
Core Equipment	**6650.00**
Network Cabling and Connectivity Components	**2253.30**
Grand Total	**8903.30**

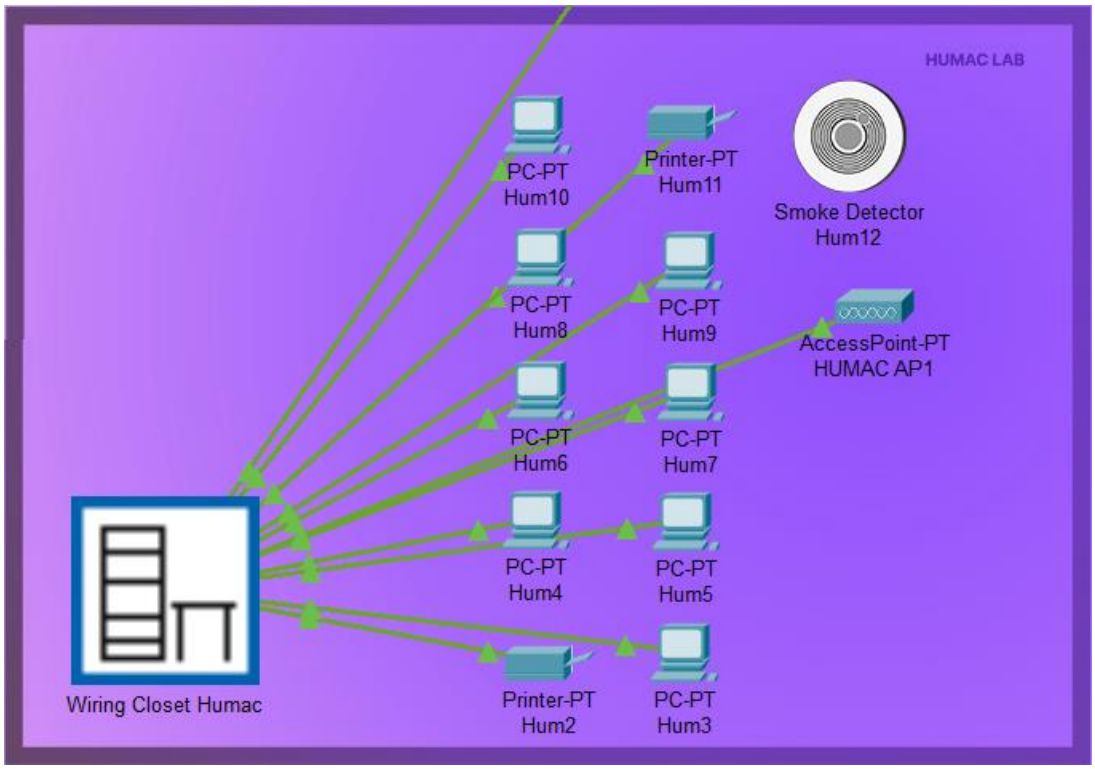
3. Physical Topology



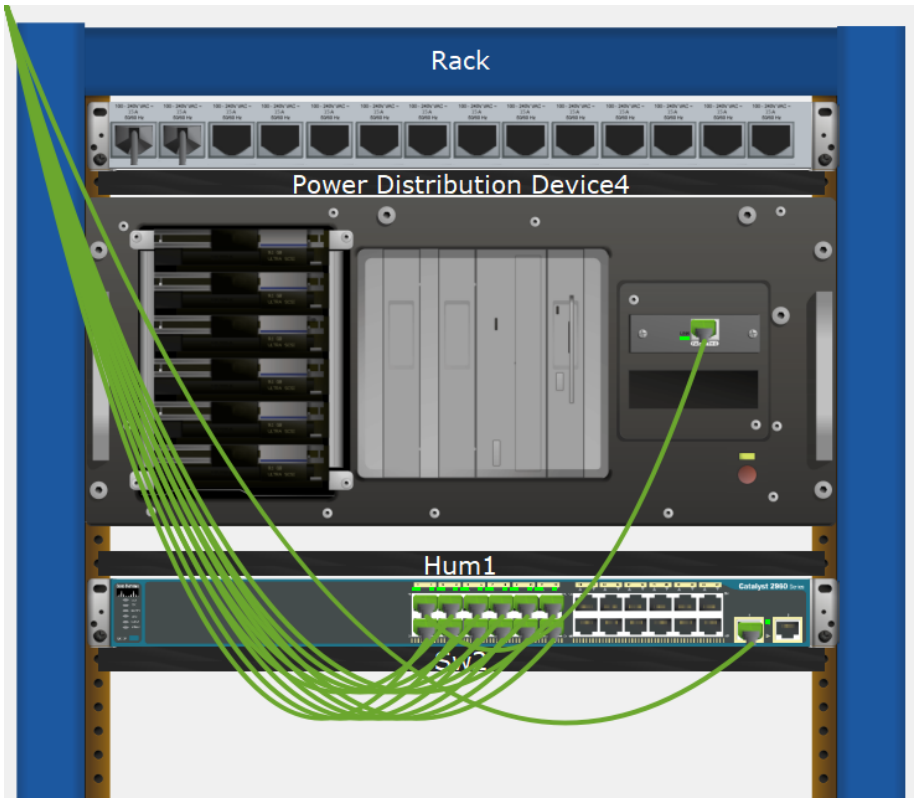
Sunway City



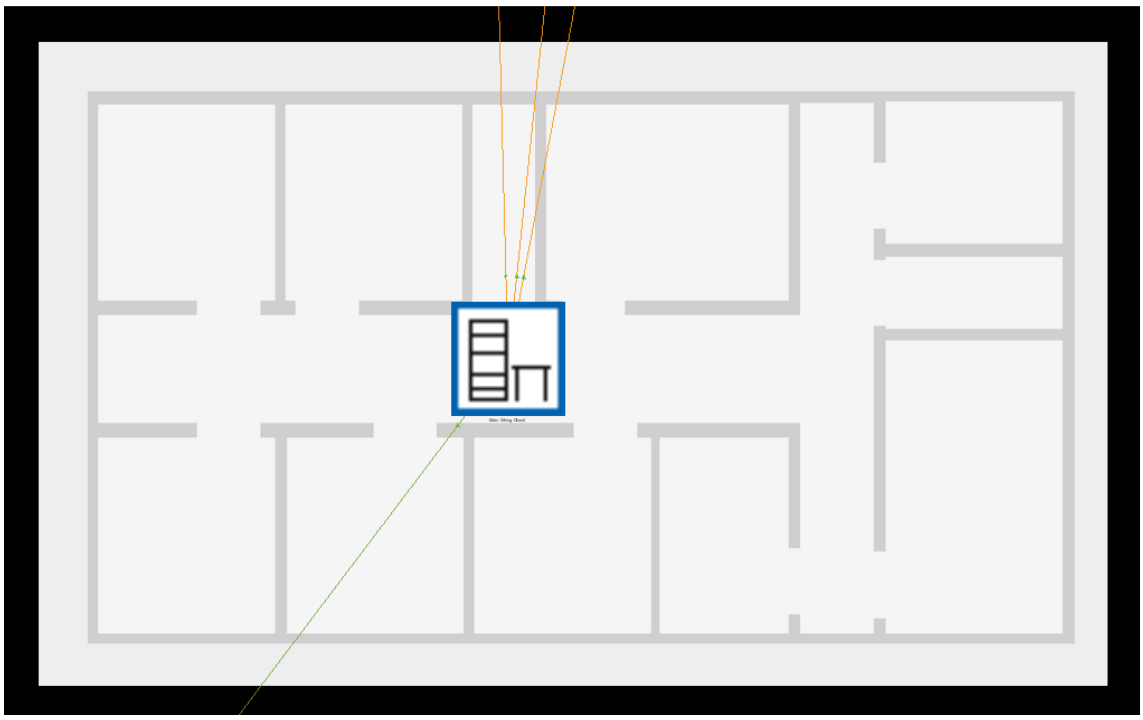
Sunway University Building - HUMAC Lab

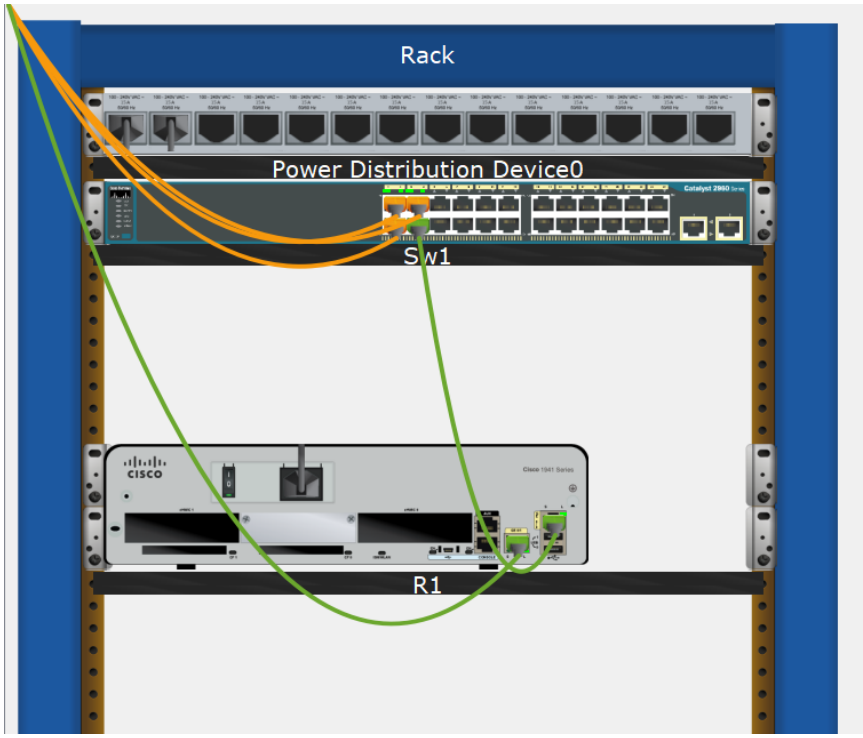


HUMAC Lab Wiring Closet

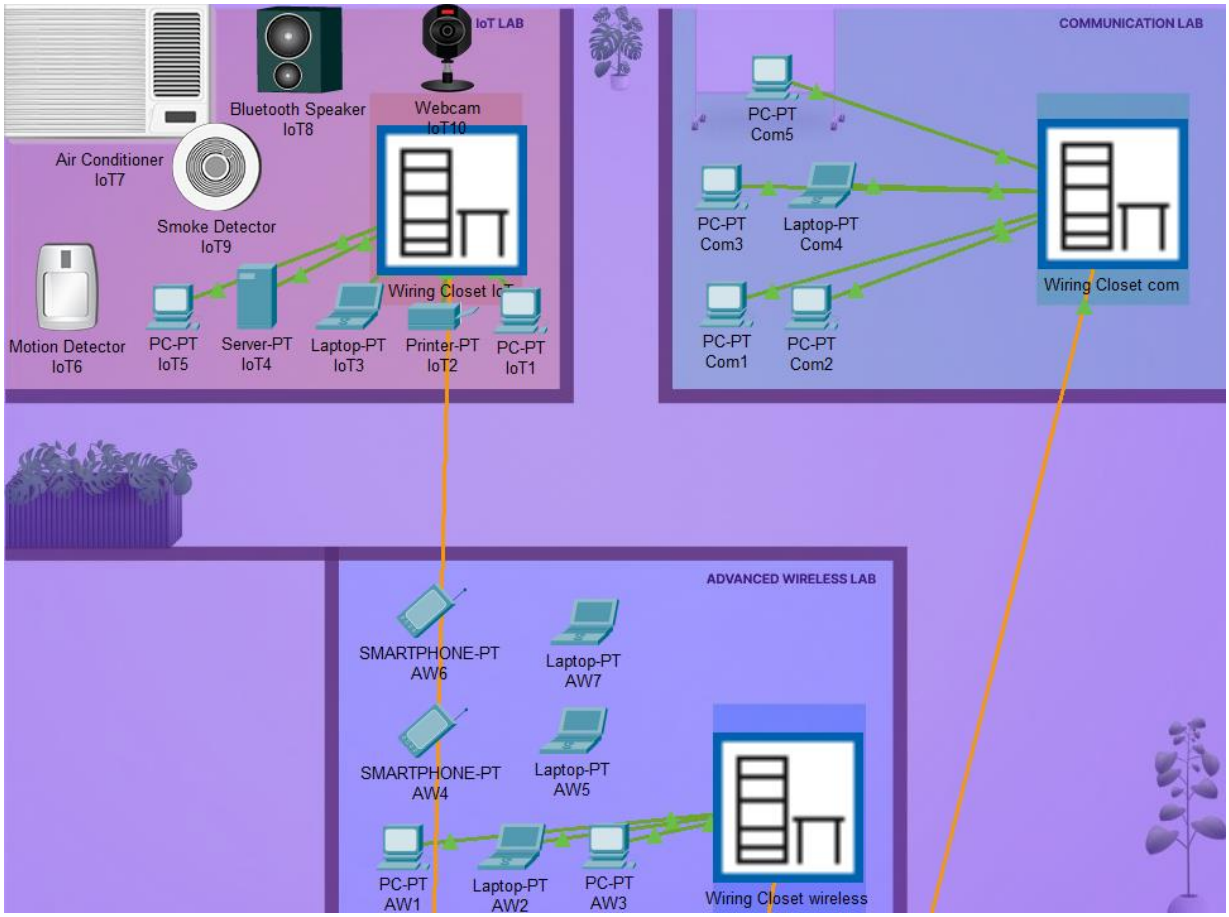


Main Wiring Closet

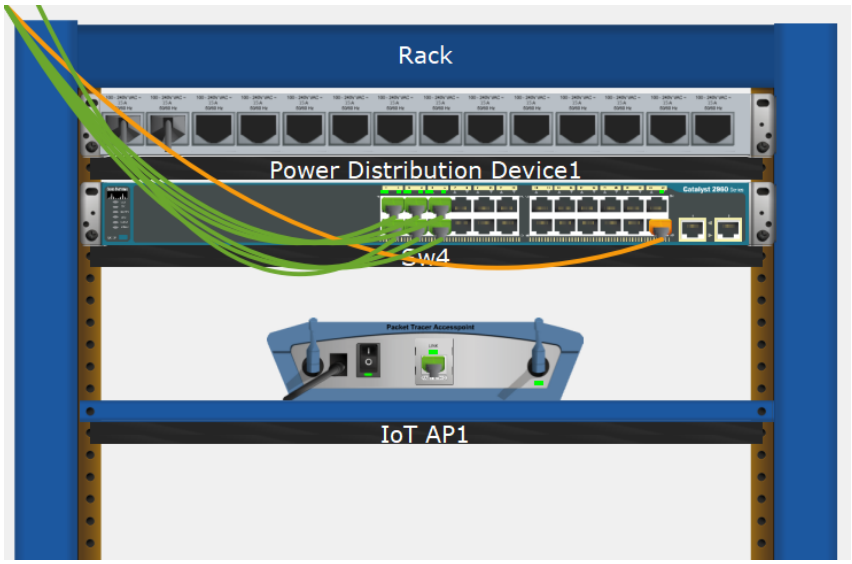




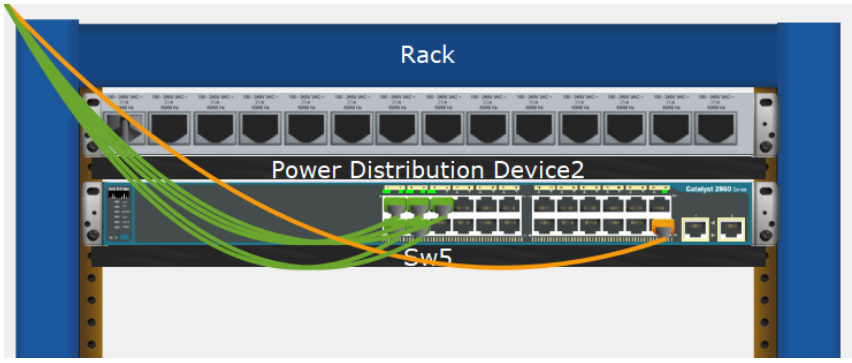
SET Building - IoT Lab, Communications Lab and Advanced Wireless Lab



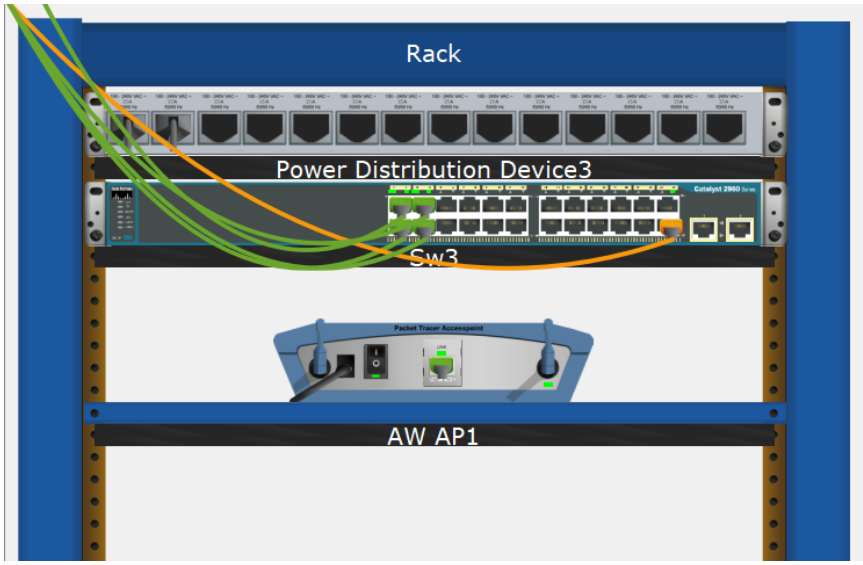
IoT Lab Wiring Closet



Communications Lab Wiring Closet

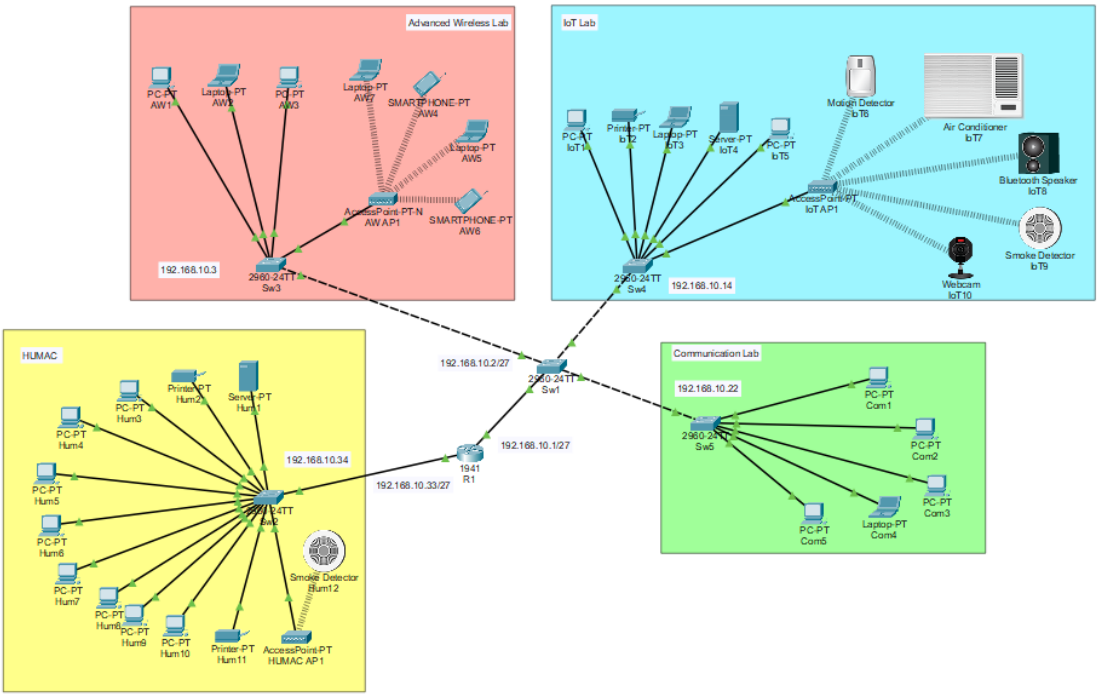


Advanced Wireless Lab Wiring Closet



4. Logical Topology

Network Address: 192.168.10.0/24



<u>Subnet Table</u>

Region	Subnet Mask	Subnet	First Usable Host Address	Last Usable Host Address	Broadcast Address
IoT Lab (VLAN1)	255.255.255.224	192.168.10.0/27	192.168.10.1	192.168.10.30	192.168.10.31
HUMAC (VLAN2)	255.255.255.224	192.168.10.32/27	192.168.10.33	192.168.10.62	192.168.10.63
Advanced Wireless Lab (VLAN3)	255.255.255.224	192.168.10.0/27	192.168.10.1	192.168.10.30	192.168.10.31
Communication Lab (VLAN4)	255.255.255.224	192.168.10.0/27	192.168.10.1	192.168.10.30	192.168.10.31

Addressing Table

Device	Interface	Address	Subnet Mask	Default Gateway
Router 1	G0/0	192.168.10.1	255.255.255.224	N/A
Router 1	G0/1	192.168.10.33	255.255.255.224	N/A
Sw1	VLAN 1	192.168.10.2	255.255.255.224	192.168.10.1
Sw2	VLAN 1	192.168.10.34	255.255.255.224	192.168.10.33
Sw3	VLAN 1	192.168.10.3	255.255.255.224	192.168.10.1
Sw4	VLAN 1	192.168.10.14	255.255.255.224	192.168.10.1
Sw5	VLAN1	192.168.10.22	255.255.255.224	192.168.10.1
HUM1	NIC	192.168.10.35	255.255.255.224	192.168.10.33
HUM2	NIC	192.168.10.36	255.255.255.224	192.168.10.33
HUM3	NIC	192.168.10.37	255.255.255.224	192.168.10.33
HUM4	NIC	192.168.10.38	255.255.255.224	192.168.10.33
HUM5	NIC	192.168.10.39	255.255.255.224	192.168.10.33
HUM6	NIC	192.168.10.40	255.255.255.224	192.168.10.33
HUM7	NIC	192.168.10.41	255.255.255.224	192.168.10.33
HUM8	NIC	192.168.10.42	255.255.255.224	192.168.10.33
HUM9	NIC	192.168.10.43	255.255.255.224	192.168.10.33
HUM10	NIC	192.168.10.44	255.255.255.224	192.168.10.33
HUM11	NIC	192.168.10.45	255.255.255.224	192.168.10.33
HUM12	NIC	192.168.10.46	255.255.255.224	192.168.10.33
COM1	NIC	192.168.10.23	255.255.255.224	192.168.10.1
COM2	NIC	192.168.10.24	255.255.255.224	192.168.10.1
COM3	NIC	192.168.10.25	255.255.255.224	192.168.10.1
COM4	NIC	192.168.10.26	255.255.255.224	192.168.10.1
COM5	NIC	192.168.10.27	255.255.255.224	192.168.10.1
AW1	NIC	192.168.10.15	255.255.255.224	192.168.10.1

AW2	NIC	192.168.10.16	255.255.255.224	192.168.10.1
AW3	NIC	192.168.10.17	255.255.255.224	192.168.10.1
AW4	NIC	192.168.10.18	255.255.255.224	192.168.10.1
AW5	NIC	192.168.10.19	255.255.255.224	192.168.10.1
AW6	NIC	192.168.10.20	255.255.255.224	192.168.10.1
AW7	NIC	192.168.10.21	255.255.255.224	192.168.10.1
IoT1	NIC	192.168.10.4	255.255.255.224	192.168.10.1
IoT2	NIC	192.168.10.5	255.255.255.224	192.168.10.1
IoT3	NIC	192.168.10.6	255.255.255.224	192.168.10.1
IoT4	NIC	192.168.10.7	255.255.255.224	192.168.10.1
IoT5	NIC	192.168.10.8	255.255.255.224	192.168.10.1
IoT6	NIC	192.168.10.9	255.255.255.224	192.168.10.1
IoT7	NIC	192.168.10.10	255.255.255.224	192.168.10.1
IoT8	NIC	192.168.10.11	255.255.255.224	192.168.10.1
IoT9	NIC	192.168.10.12	255.255.255.224	192.168.10.1
IoT10	NIC	192.168.10.13	255.255.255.224	192.168.10.1

Part B: Implementation

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>enable
Router#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R1
R1(config)#line config 0
                ^
% Invalid input detected at '^' marker.

R1(config)#li
R1(config)#line ?
  <2-499>  First Line number
  aux      Auxiliary line
  console  Primary terminal line
  tty      Terminal controller
  vty      Virtual terminal
  x/y/z    Slot/Subslot/Port for Modems
R1(config)#line co
R1(config)#line console 0
R1(config-line)#password sunway123
R1(config-line)#login
R1(config-line)#exit
R1(config)#enable secret sunwayedu
R1(config)#banner m
R1(config)#banner motd "Authorized Access Only!!"
R1(config)#in
R1(config)#interface gig
R1(config)#interface gigabitEthernet 0/0
R1(config-if)#ip address 192.168.10.1 255.255.255.224
R1(config-if)#no shu
R1(config-if)#no shutdown
R1(config-if)#des
R1(config-if)#description Connection to Sw1
R1(config-if)#exit
R1(config)#inte
R1(config)#interface gig
R1(config)#interface gigabitEthernet 0/1
R1(config-if)#ip address 192.168.10.33 255.255.255.224
R1(config-if)#no sh
R1(config-if)#no shutdown
R1(config-if)#descr
R1(config-if)#description Connection to Sw2
R1(config-if)#exit
R1(config)#se
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

Top

Sw1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up


Switch>enable
Switch#config term
Switch#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hos
Switch(config)#hostname Sw1
Sw1(config)#enable s
Sw1(config)#enable secret sunwayedu
Sw1(config)#line console 0
Sw1(config-line)#password sunway123
Sw1(config-line)#login
Sw1(config-line)#exit
Sw1(config)#line vty 1
Sw1(config-line)#passowrd sunway123
                    ^
% Invalid input detected at '^' marker.

Sw1(config-line)#password sunway123
Sw1(config-line)#login
Sw1(config-line)#exit
Sw1(config)#interfa
Sw1(config)#interface vlan 1
Sw1(config-if)#ip address 192.168.10.2 255.255.255.224
Sw1(config-if)#no shutdown
Sw1(config-if)#desc
Sw1(config-if)#description Connection to R1
Sw1(config-if)#exit
Sw1(config)#ip defau
Sw1(config)#ip default-gateway 192.168.10.1
Sw1(config)#bann
Sw1(config)#banner motd "Authorized Access Only!!"
Sw1(config)#service passwo
Sw1(config)#service password-encryption
Sw1(config)#exit
Sw1#
%SYS-5-CONFIG_I: Configured from console by console

Sw1#exit
```

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Sw2

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>enable
Switch#config termi
Switch#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hos
Switch(config)#hostname Sw2
Sw2(config)#line con
Sw2(config)#line console 0
Sw2(config-line)#password sunway123
Sw2(config-line)#login
Sw2(config-line)#exit
Sw2(config)#line vy
Sw2(config)#line vt
Sw2(config)#line vty 1
Sw2(config-line)#pass
Sw2(config-line)#password sunway123
Sw2(config-line)#exit
Sw2(config)#enable secr
Sw2(config)#enable secret sunwayedu
Sw2(config)#line vt
Sw2(config)#line vty 1
Sw2(config-line)#login
Sw2(config-line)#exit
Sw2(config)#confi
Sw2(config)#in
Sw2(config)#interface vt
Sw2(config)#interface v
Sw2(config)#interface vlan 1
Sw2(config-if)#ip add
Sw2(config-if)#ip address 192.168.10.34 255.255.255.224
Sw2(config-if)#no sh
Sw2(config-if)#no shutdown
Sw2(config-if)#des
Sw2(config-if)#description Connection to R1
Sw2(config-if)#exit
Sw2(config)#service pass
Sw2(config)#service password-encryption
Sw2(config)#ip defau
Sw2(config)#ip default-gateway 192.168.10.33
Sw2(config)#pas
Sw2(config)#bann
Sw2(config)#banner motd "Authorized Access Only!!"
Sw2(config)#exit
Sw2#
%SYS-5-CONFIG_I: Configured from console by console
```

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Sw3

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up


Switch>enable
Switch#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Sw3
Sw3(config)#enable secret sunwayedu
Sw3(config)#line console 0
Sw3(config-line)#password sunway123
Sw3(config-line)#login
Sw3(config-line)#exit
Sw3(config)#line vty 1
Sw3(config-line)#password sunway123
Sw3(config-line)#login
Sw3(config-line)#exit
Sw3(config)#service pass
Sw3(config)#service password-encryption
Sw3(config)#banner motd "Authorized Access Only!!"
Sw3(config)#inter
Sw3(config)#interface vlan 1
Sw3(config-if)#ip address 192.168.10.3 255.255.255.224
Sw3(config-if)#no shutdown
Sw3(config-if)#description Connection to Sw1
Sw3(config-if)#exit
Sw3(config)#ip defaul
Sw3(config)#ip default-gateway 192.168.10.1
Sw3(config)#
```

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Sw4

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up


Switch>enable
Switch#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostn
Switch(config)#hostname Sw4
Sw4(config)#enable secret sunwayedu
Sw4(config)#line console 0
Sw4(config-line)#password sunway123
Sw4(config-line)#login
Sw4(config-line)#exit
Sw4(config)#banner motd "Authorized Access Only!!"
Sw4(config)#ip defau
Sw4(config)#ip default-gateway 192.168.10.1
Sw4(config)#service pass
Sw4(config)#service password-encryption
Sw4(config)#in
Sw4(config)#interface vlan 1
Sw4(config-if)#ip address 192.168.10.14 255.255.255.224
Sw4(config-if)#no shutdown
Sw4(config-if)#description "Connection to Sw1"
Sw4(config-if)#exit
Sw4(config)#line vty 1
Sw4(config-line)#password sunway123
Sw4(config-line)#login
Sw4(config-line)#exit
Sw4(config)#service pass
Sw4(config)#service password-encryption
Sw4(config)#
```

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Sw5

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up


Switch>enable
Switch#config  terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Sw5
Sw5(config)#enable secret sunwayedu
Sw5(config)#line console 0
Sw5(config-line)#password sunway123
Sw5(config-line)#login
Sw5(config-line)#exit
Sw5(config)#line vty 1
Sw5(config-line)#password sunway123
Sw5(config-line)#login
Sw5(config-line)#exit
Sw5(config)#service pass
Sw5(config)#service password-encryption
Sw5(config)#ip defaul
Sw5(config)#ip default-gateway 192.168.10.1
Sw5(config)#inter
Sw5(config)#interface vlan 1
Sw5(config-if)#ip address 192.168.10.22 255.255.255.224
Sw5(config-if)#no shutdown
Sw5(config-if)#description Connection to Sw1
Sw5(config-if)#exit
Sw5(config)#banner motd "Authorized Access Only!!"
Sw5(config)#
```

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Part C: Lessons Learned

Teamwork Experience

Our team collaborated effectively to achieve both the design and implementation goals. We divided tasks based on individual strengths, with one member focusing on the network design, another on subnetting and IP addressing, and others on device configuration and testing. Regular meetings ensured everyone was aligned, and we used tools in Microsoft word to track progress and assign tasks. Communication was key, we held daily stand-ups to discuss challenges and progress. To enhance teamwork, we encouraged open feedback and maintained a positive attitude, which helped us stay motivated and resolve conflicts quickly.

Technical Challenges

Several technical challenges arose during the project:

1. **Subnetting Errors:** Initially, we miscalculated the subnet ranges, leading to IP conflicts. We resolved this by thoroughly reading the slides and double-checking our calculations.
2. **Router Configuration:** Misconfigurations in the router interfaces caused connectivity issues between subnets. We referred to Cisco documentation and tested each interface step-by-step to fix the problem.
3. **Device Connectivity:** Some PCs in the IoT Lab couldn't communicate with others due to incorrect switch configurations. We verified the switch settings and ensured VLANs were properly configured.

Issue Resolution

To address these challenges, we employed a systematic approach:

1. **Problem Identification:** We used the **divide-and-conquer** method to isolate issues, starting from the physical layer and moving up the OSI model.
2. **Collaborative Troubleshooting:** Team members with stronger technical skills guided others through complex configurations, ensuring everyone learned in the process.
3. **Documentation:** We maintained detailed logs of configurations and changes, which helped us backtrack and identify the root cause of issues quickly.
4. **Testing:** After each fix, we tested connectivity.

   using **ping** and **traceroute** commands to ensure the network was functioning as intended.

Overall Experience

This project was a valuable hands-on learning experience. We gained practical skills in network design, subnetting, device configuration, and troubleshooting. The most valuable takeaway was understanding how theoretical concepts apply in real-world scenarios, such as the importance of proper IP addressing and router configuration. We also learned the significance of teamwork and communication in achieving complex technical goals. This project reinforced the importance of hands-on learning in networking, as it allowed us to apply classroom knowledge to solve real problems. Overall, the experience has prepared us for future networking challenges and deepened our appreciation for the field.